

TØ 11 – Mikroøkonomi 2 – 2024

Problems for Tutorial

Problem 1

Assume that Donald wants to contract a cheerleader to work for him in his new government. A cheerleader in his government exerts verifiable effort e and gets wage w . Donald is risk-neutral and values cheerleader effort by $\Pi(e, w) = ke - w$. k is large enough to ensure that Donald is interested in contracting any cheerleader $e > 0$. The cheerleader is risk neutral and is either of type 1 with utility function $U^1 = w - e^2$ or type 2 with utility function $U^2 = w - 2e^2$. The probability the cheerleader is of type 1 is q and thus the probability that he is of type 2 is $(1 - q)$. Both types of cheerleaders have reservation utility $\underline{u} = 0$.

1. Formulate Donald's problem under the first best (when Donald observes what type of cheerleader he is contracting with).
2. Is the first best implementable when Donald does not observe the type of cheerleader (second best)? Why/Why not?
3. Formulate Donald's problem under the second best. Argue which constraints are binding and non-binding.
4. Solve the problem calculating the optimal contract characteristics under the second best.
5. Compare first and second best contracts. Explain any differences.
6. Assume that Donald's utility is instead $\Pi(e, w) = e - w$ and $q = \frac{1}{2}$. Calculate contracts under the first and second best.

Profit is

$$\Pi(e_i, w) = ke_i - w$$

Utility of the workers are

$$u_G = w_G - e_G^2$$

$$u_B = w_B - 2e_B^2$$

Probabilities

$$P(\text{Type } G) = q$$

$$P(\text{Type } B) = 1 - q$$

Reservation utility is

$$\bar{u} = 0$$

1

Principal's maximisation problem under first best with the **type G** b (cf. the slideshow "Adverse selection - part 2")

$$\begin{aligned} \max & ke - w \\ \text{s.t.} & \\ & w_G - e_G^2 \geq 0 \end{aligned}$$

Using Lagrange:

$$\mathcal{L} = ke - w_G + \lambda(w_G - e^2)$$

The FOCs are

$$\frac{\partial \mathcal{L}}{\partial e} = k - 2\lambda e = 0 \Leftrightarrow \lambda = \frac{1}{2e}k$$

$$\frac{\partial \mathcal{L}}{\partial w_G} = -1 + \lambda = 0 \Leftrightarrow \lambda = 1$$

Because $\lambda = 1$,

$$\lambda = \frac{1}{2e}k \Leftrightarrow e = \frac{1}{2\lambda}k = \frac{1}{2}k$$

Implying that

$$w_G - e^2 = w_G - \left(\frac{1}{2}k\right)^2 = w_G - \frac{1}{4}k^2 = 0 \Leftrightarrow w_G = \frac{1}{4}k^2$$

The optimal pair is therefore

$$(e_G, w_G) = \left(\frac{1}{2}k, \frac{1}{4}k^2\right)$$

Now, **type B**

$$\begin{aligned} \max & ke - w \\ \text{s.t.} & \\ & w_B - 2e_B^2 \geq 0 \end{aligned}$$

Using Lagrange:

$$\mathcal{L} = ke - w_B + \lambda(w_B - 2e^2)$$

The FOCs are

$$\frac{\partial \mathcal{L}}{\partial e} = k - 4\lambda e = 0 \Leftrightarrow \lambda = \frac{1}{4e}k$$

$$\frac{\partial \mathcal{L}}{\partial w_B} = -1 + \lambda = 0 \Leftrightarrow \lambda = 1$$

Because $\lambda = 1$,

$$\lambda = \frac{1}{4e}k \Leftrightarrow e = \frac{1}{4\lambda}k = \frac{1}{4}k$$

Implying that

$$w_B - 2e_B^2 = w_B - 2\left(\frac{1}{4}k\right)^2 = w_B - 2\frac{1}{16}k^2 = w_B - \frac{1}{8}k^2$$

The optimal pair is therefore

$$(e_B, w_B) = \left(\frac{1}{4}k, \frac{1}{8}k^2\right)$$

2

No, the first best is not implementable since type G will pretend to be type B and thus earning a higher wage (c.f. page 7 in the same slideshow as before)

This is the high type's utility function with the values found in part 1: $(e_G, w_G) = \left(\frac{1}{2}k, \frac{1}{4}k^2\right)$

$$u_G(w_G, e_G) = w_G - e_G^2 = \frac{1}{4}k^2 - \left(\frac{1}{2}k\right)^2 = \frac{1}{4}k^2 - \frac{1}{4}k^2 = 0 = \bar{u}$$

If he pretends to be type B, he can instead get the following:

$$u_G(w_B, e_B) = w_B - e_B^2 = \frac{1}{8}k^2 - \left(\frac{1}{4}k\right)^2 = \frac{1}{8}k^2 - \frac{1}{16}k^2 = \frac{1}{16}k^2$$

Since $(e_B, w_B) = \left(\frac{1}{4}k, \frac{1}{8}k^2\right)$

Meaning that

$$u_G(w_B, e_B) > u_G(w_G, e_G)$$

3

$$\max q(ke_G - w_G) + (1 - q)(ke_B - w_B)$$

s. t.

$$w_G - e_G^2 \geq \bar{u} = 0$$

$$w_B - 2e_B^2 \geq \bar{u} = 0$$

$$w_G - e_G^2 \geq w_B - e_B^2$$

$$w_B - 2e_B^2 \geq w_G - 2e_G^2$$

The constraints are PC_G, PC_B, IC_G and IC_B

4

Because $PC_G \geq PC_B \geq 0$, we can omit the participation constraint for type G, i.e., PC_G . This implies that **only PC_B and IC_G bind.**

Therefore, the constraints are now

$$w_B - 2e_B^2 = 0 \Leftrightarrow w_B = 2e_B^2$$

$$w_G - e_G^2 = w_B - e_B^2 \Leftrightarrow w_G = w_B - e_B^2 + e_G^2 = 2e_B^2 - e_B^2 + e_G^2$$

Where the last equality is because

$$w_B = 2e_B^2$$

Because of the binding PCB

The maximisation problem is now

$$\begin{aligned} \max & q(ke_G - w_G) + (1 - q)(ke_B - w_B) \\ \text{s. t. } & w_B - 2e_B^2 \geq 0 \\ \text{s. t. } & w_G - e_G^2 \geq w_B - e_B^2 \end{aligned}$$

Lagrangian

$$\mathcal{L} = q(ke_G - w_G) + (1 - q)(ke_B - w_B) + \lambda(w_B - 2e_B^2) + \mu(w_G - e_G^2 - w_B + e_B^2)$$

FOCs

$$\begin{aligned} w_G: & q = \mu \\ e_G: & qk = 2e_G\mu \Leftrightarrow e_G = \frac{qk}{2\mu} = \frac{qk}{2q} = \frac{1}{2}k \\ w_B: & -(1 - q) = -\lambda + \mu \Leftrightarrow \lambda = 1 - q + \mu = 1 - q + q = 1 > 0 \\ e_B: & (1 - q)k = 4e_B\lambda - 2e_B\mu = 4e_B - 2e_Bq = e_B(4 - 2q) \Leftrightarrow e_B = \frac{(1 - q)k}{4 - 2q} \end{aligned}$$

Where in the last FOC, I have used $\lambda = 1$ and $\mu = q$

The optimal pairs are now found using the binding PCB and ICG:

$$\begin{aligned} w_B = 2e_B^2 & \Leftrightarrow w_B = 2\left(\frac{(1 - q)k}{4 - 2q}\right)^2 \\ w_G = e_G^2 + w_B - e_B^2 & \Leftrightarrow \left(\frac{1}{2}k\right)^2 + 2\left(\frac{(1 - q)k}{4 - 2q}\right)^2 - \left(\frac{(1 - q)k}{4 - 2q}\right)^2 = \left(\frac{1}{2}k\right)^2 + \left(\frac{(1 - q)k}{4 - 2q}\right)^2 \end{aligned}$$

5

In FB, the contracts were

$$\begin{aligned} (e_G, w_G) &= \left(\frac{1}{2}k, \frac{1}{4}k^2\right) \\ (e_B, w_B) &= \left(\frac{1}{4}k, \frac{1}{8}k^2\right) \end{aligned}$$

And in SB

$$(e_G^{SB}, w_G^{SB}) = \left(\frac{1}{2}k, \left(\frac{1}{2}k \right)^2 + \left(\frac{(1-q)k}{4-2q} \right)^2 \right)$$

$$(e_B^{SB}, w_B^{SB}) = \left(\frac{(1-q)k}{4-2q}, 2 \left(\frac{(1-q)k}{4-2q} \right)^2 \right)$$

Good type:

Notice that Donald pays a higher wage to the good type compared to in FB since he wants to make sure she does not pretend to be the bad type.

$$w_G^{SB} > w_G^{FB} \Leftrightarrow \left(\frac{1}{2}k \right)^2 + \left(\frac{(1-q)k}{4-2q} \right)^2 > \frac{1}{4}k^2 \Leftrightarrow \left(\frac{(1-q)k}{4-2q} \right)^2 > 0$$

He requires same level of effort since the cheerleader is risk-neutral: $e_G^{FB} = e_G^{SB} = \frac{1}{2}k$ (see slides)

Bad type:

Donald requires less effort from the bad type in SB since the contract is distorted because Donald must ensure that the good type will not pretend to be the bad type

$$e_B^{FB} > e_B^{SB} \Leftrightarrow \frac{1}{4}k > \frac{(1-q)k}{4-2q} \Leftrightarrow 2q > 0$$

And Donald therefore pays a lower wage:

$$w_B^{FB} > w_B^{SB} \Leftrightarrow \frac{1}{8}k^2 > 2 \left(\frac{(1-q)k}{4-2q} \right)^2 \Leftrightarrow 0 < q < \frac{4}{3}$$

6

First best:

From part 1, we had

$$(e_G, w_G) = \left(\frac{1}{2}k, \frac{1}{4}k^2 \right)$$

$$(e_B, w_B) = \left(\frac{1}{4}k, \frac{1}{8}k^2 \right)$$

Inserting $k = 1$ yields

$$E[\pi] = q(e_G - w_G) + (1-q)(e_B - w_B) = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{4} \right) + \frac{1}{2} \left(\frac{1}{4} - \frac{1}{8} \right) = \frac{3}{16} = 0.1875$$

Second best (just insert $k = 1, q = \frac{1}{2}$ in the expressions for w_B, w_G from part 5):

$$(w_G^{SB}, e_G^{SB}) = \left(\frac{5}{36}, \frac{1}{2}\right)$$

$$(w_B^{SB}, e_B^{SB}) = \left(\frac{1}{18}, \frac{1}{6}\right)$$

Inserting $k = 1$ yields

$$E[\pi] = q(e_G - w_G) + (1 - q)(e_B - w_B) = \frac{1}{2}\left(\frac{1}{2} - \frac{5}{36}\right) + \frac{1}{2}\left(\frac{1}{6} - \frac{1}{18}\right) \approx 0.264$$

So, profits are lower under FB

7. As in part 6, assume that Donald's utility is $\Pi(e, w) = e - w$ and $q = \frac{1}{2}$. Consider the case where in the second best Donald only offer a contract to type 1 and thus foregoes the opportunity of contracting with type 2 also. What contract does Donald offer and is it worthwhile?

7

Donald implements FB contract to type 1 (Good type):

$$(e_G, w_G) = \left(\frac{1}{2}k, \frac{1}{4}k^2\right) = \left(\frac{1}{2}, \frac{1}{4}\right)$$

Profits are then

$$q(e - w) = q(e_G - w_G) = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{1}{8} < 0.264$$

Where 0.264 is from usual SB (part 6) so not a good idea. He should stick with the two contracts in SB instead of only offering a good type contract

Problem 2

A small IT firm is owned by a risk-averse current owner (A) with initial wealth W_0 and an increasing, strictly concave von Neumann-Morgenstern utility function $u_E(w)$ where w is final wealth (i.e. including initial wealth, the value of the firm and any transfers) of the individual. The current owner is considering selling (a part of) his firm.

There is only one potential buyer, Googles (P). Googles acts like a monopolistic risk neutral buyer and proposes a contract consisting of (x, T) where $x \in [0, 1]$ is the share of the firm the current owner keeps (he thus sells the share $1 - x$ to Googles) and $T \geq 0$ is the payment to the current owner.

The value of the firm is given by $\theta + \gamma$ where θ and γ are two independent random variables. The value of θ is known to its current owner but not to anyone else. It is however known to all players that θ can take two values θ_L and θ_H with probability β and $1 - \beta$ respectively and $\theta_L < \theta_H$. γ is a continuous random variable with expected value is 0. γ is only realized in the future (i.e. after a potential trade of the firm) and is thus unknown to everybody at the time of purchase.

The timing of the game is as follows:

1. Nature chooses θ .
2. Googles proposes the terms of trade (i.e. offers a contract)
3. The current owner accepts or rejects the contract.

If the current owner rejects the contract he gets the full value of the firm, if he accepts the contract he gets a transfer T and a share of the value of the firm x . We can therefore express the expected utility of a current owner of type θ_i associated with a given contract (x, T) by

$$E(u_E(w)) = \int u(x \cdot (\theta_i + \gamma) + T + W_0) f(\gamma) d\gamma$$

Under the first best both the current owner and Googles know the type θ .

Questions:

1. Formulate the participation constraint and discuss the outside option's dependence on type.

1

Principal (Google) will maximise their own part of the firm times the value minus the price of the takeover, T . Notice that γ is excluded from the max since it is unknown at the time of the purchase

We do not need the β and $1 - \beta$ here as this is FB—just like we did not need the q and $1 - q$ in the previous problem in FB since they are directly observable

Maximisation problem is: (with $i \in \{L, H\}$):

$$\max(1 - x)\theta_i - T_i$$

St

$$\int u(x(\theta_i + \gamma) + T + W_0) f(\gamma) d\gamma \geq \int u((\theta_i + \gamma) + W_0) f(\gamma) d\gamma = \bar{u}$$

2. Explain why the PC constraint is binding under the first best and why we expect Googles to offer a contract where $x = 0$ (2-3 sentences).
3. (c) Formulate the first best problem. Show that the first order conditions imply efficient risk sharing. Explain the intuition (2 sentences).
4. Are the first best contracts incentive compatible under the second best?
5. Write down the second best problem and assume that the second best contract will have $x_L = 0$. Go through the different constraints and discuss which of them we can expect to be binding.
6. Identify and explain the central trade-off between rent extraction and risk premiums which the second best problem describes.

2

PC binds since Google is a monopolist so it can decrease T until PC is binding if it does not bind at first. Also, as we have a risk averse agent and a risk neutral principle, the cheapest way to satisfy the PC would be to offer $x = 0$. The intuition is that since the agent is risk-averse, he will accept to sell all his shares of the firm, leaving him with $x = 0$

3

Lagrange

$$\mathcal{L} = (1 - x)\theta_i - T_i + \lambda \left(\int u(x(\theta_i + \gamma) + T + W_0)f(\gamma)d\gamma - \int u((\theta_i + \gamma) + W_0)f(\gamma)d\gamma \right)$$

FOCs

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial T} &= -1 + \lambda \int u'(x(\theta_i + \gamma) + T + W_0)f(\gamma)d\gamma = 0 \\ \Leftrightarrow \lambda &= \frac{1}{\int [u'(x(\theta_i + \gamma) + T + W_0)]f(\gamma)d\gamma} \end{aligned}$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= -\theta_i + \lambda \int u'(x(\theta_i + \gamma) + T + W_0)(\theta_i + \gamma)f(\gamma) d\gamma = 0 \Leftrightarrow 0 \\ &= -\theta_i + \frac{1}{\int u'(x(\theta_i + \gamma) + T + W_0)f(\gamma)d\gamma} \cdot \int u'(x(\theta_i + \gamma) + T + W_0)(\theta_i + \gamma)f(\gamma) d\gamma \end{aligned}$$

$$\begin{aligned}
&= -\theta_i + \frac{\int u'(x(\theta_i + \gamma) + T + W_0)(\theta_i + \gamma)f(\gamma) d\gamma}{\int u'(x(\theta_i + \gamma) + T + W_0)f(\gamma)d\gamma} \\
&= -\theta_i + \frac{\int (\theta_i + \gamma)f(\gamma)d\gamma}{\int f(\gamma)d\gamma} \\
&= -\theta_i + \frac{\int \theta_i f(\gamma)d\gamma}{\int f(\gamma)d\gamma} + \frac{\int \gamma f(\gamma)d\gamma}{\int f(\gamma)d\gamma} \\
&= -\theta_i + \theta_i + \frac{\mathbb{E}[\gamma]}{\int f(\gamma)d\gamma} \\
&= -\theta_i + \theta_i
\end{aligned}$$

Since $\mathbb{E}[\gamma] = 0$

So, $0 = 0$ meaning that $x = 0$

Second line comes from the value of λ found in the other FOC

4

Since for both types, $x = 0$, the low type will pretend to be the high type and hence, get a higher transfer, T

5

$$\max \beta((1 - x_L)(\theta_L + \gamma) - T_L) + (1 - \beta)((1 - x_H)(\theta_H + \gamma) - T_H)$$

St

PCL, PCH, ICL and ICH

$$\begin{aligned}
&\int u(x_L(\theta_L + \gamma) + T_L + W_0)f(\gamma)d\gamma \geq \bar{u}_L \\
&\int u(x_H(\theta_H + \gamma) + T_H + W_0)f(\gamma)d\gamma \geq \bar{u}_H \\
&\int u(x_L(\theta_L + \gamma) + T_L + W_0)f(\gamma)d\gamma \geq \int u(x_H(\theta_L + \gamma) + T_H + W_0)f(\gamma)d\gamma \\
&\int u(x_H(\theta_H + \gamma) + T_H + W_0)f(\gamma)d\gamma \geq \int u(x_L(\theta_H + \gamma) + T_L + W_0)f(\gamma)d\gamma
\end{aligned}$$

The two PCs are binding—they can both bind because of these different reservation utilities

ICL binds since FB is not implementable:

$$\int u(x_L(\theta_L + \gamma) + T_L + W_0)f(\gamma)d\gamma = \int u(x_H(\theta_H + \gamma) + T_H + W_0)f(\gamma)d\gamma$$

ICH is non-binding since $x_L = 0$ implies:

$$\int u(x_L(\theta_L + \gamma) + T_L + W_0)f(\gamma)d\gamma = \int u(x_L(\theta_H + \gamma) + T_L + W_0)f(\gamma)d\gamma$$

And because $\theta_L < \theta_H$,

$$\int u(x_H(\theta_H + \gamma) + T_H + W_0)f(\gamma)d\gamma > \int u(x_H(\theta_L + \gamma) + T_H + W_0)f(\gamma)d\gamma$$

So,

$$\int u(x_H(\theta_L + \gamma) + T_H + W_0)f(\gamma)d\gamma > \int u(x_L(\theta_L + \gamma) + T_L + W_0)f(\gamma)d\gamma$$

Since $T_H > T_L$

Implying that the ICH does not bind:

$$\int u(x_H(\theta_H + \gamma) + T_H + W_0)f(\gamma)d\gamma > \int u(x_L(\theta_H + \gamma) + T_L + W_0)f(\gamma)d\gamma$$

To sum up: PCL, PCH and ICL bind. ICH does not

6

Since the outside option varies across types: $\bar{u}_L \neq \bar{u}_H$ —unlike in a usual screening problem—the PCL cannot be omitted

The trade-off is that by increasing the risk premium of high type, x_H . At the same time, however, this makes it more costly for the low type to accept the high type's contract. To sum up, P can be surer that type L does not pretend to be type H by increasing x_H , but it costs.